Who Gets Remembered?

Gender, scientific impact and visibility on scientists' Wikipedia pages

Project plan and roadmap · Course 02805, DTU · Version 1, 11 November 2026

1. What we will do

We will build a network of scientists' English Wikipedia pages, linked by the hyperlinks between them, and add each scientist's gender (from Wikidata) and scientific impact (citations from OpenAlex). We then ask whether Wikipedia gives women scientists less visibility than their impact would predict, both in the link network (how central their pages are) and in the text (how their pages describe them). The key design choice is to compare scientists **of similar impact**, so that a gap cannot simply be explained by differences in citations.

Research question

Are women scientists less central in Wikipedia's link network, and described differently on their pages, than men with similar scientific impact?

Sub-questions

- **RQ1 · Network:** At equal citations, do women's pages have lower in-degree and PageRank than men's?
- **RQ2 · Text:** At equal citations, do women's pages use more personal and family language (e.g. *married*, *husband*, *children*) and less language about their work?
- **RQ3 · Structure:** Who links to women's pages? Are links assortative by gender, and do women cluster in particular communities or subfields?

What would change or end the question

- If the gap disappears once we compare people of similar impact, our answer to RQ1 is “no”: the inequality sits upstream, in citations, not in Wikipedia. This is a valid result, so we test it **first** (milestone M2) to leave time to reframe.
- If fewer than about 100 women can be matched to reliable impact data, the comparison is too weak and we switch to a fallback (Section 8).

2. Data and scope

Scope (to confirm in Phase 0): physicists and chemists with an English Wikipedia page, born 1920–1980, so that most of their work falls in the period OpenAlex covers well. Target size: **1,500–3,000 nodes**, within the course's recommended range. A second field (e.g. biology) is kept for the robustness check.

Source	What we take	Role in the project	Access
Wikidata	The list of scientists; gender (P21), occupation (P106), birth year, and identifiers such as ORCID or OpenAlex ID where available	Defines who is in the dataset and gives labels we don't have to guess	SPARQL query service (free)
English Wikipedia	Page text and hyperlinks between scientists' pages	The network half and the text half	MediaWiki API, with pauses (as for Marvel)
OpenAlex	Total citations, number of works, first publication year	Independent measure of scientific impact	Open REST API (free)

Network: nodes = scientists' pages; directed edges = a hyperlink from one page to another within the set. Node attributes: gender, field, birth year, citations, works.
Text: one document per page (plain text, references removed).
Data log: we record every query, its date and the number of records returned.

3. Analysis plan and checks

Following the course rule that every analysis needs a check and a sentence on what it captures and what it misses:

Analysis	Captures	Misses	Check
In-degree and PageRank vs. citations, by gender (RQ1)	Visibility within Wikipedia relative to scientific impact	Why links exist; links reflect editors and topic, not merit	Shuffle gender labels within citation bins (1,000 runs, z-score); repeat in a second field
Gender assortativity (RQ3)	Whether pages link mostly to same-gender pages	Field composition can force this pattern	Degree-preserving shuffle; label shuffle within fields
Louvain communities (RQ3)	Subfields or groups where women concentrate	Communities are not facts; they vary by seed	Several seeds with NMI; Q compared with shuffled network
Personal vs. professional wording (RQ2)	How much of a page is about private life	Word lists miss context and phrasing	Hand-label ~150 sentences; report precision and recall of the lists
TF-IDF + Scattertext, women vs. men (RQ2)	Words that distinguish the two groups	Names and field words dominate unless filtered	Name filter; compare within impact bins
Page length vs. citations, by gender (RQ2)	How much is written about each person	Length depends on editor attention, era, field	Same within-bin comparison as RQ1

4. Planned figures (three)

#	Figure	Answers
1	Scatter of citations (x, log) vs. PageRank (y, log), colored by gender, with a trend line per group; inset: null distribution of the gap with the observed value marked	RQ1
2	Scattertext-style comparison of women's vs. men's pages, plus a bar chart of the share of personal-life sentences by gender within impact bins	RQ2
3	The scientist network, colored by gender and grouped by community, with node size by citations	RQ3

5. Roadmap

Project sessions are on 11, 18 and 25 November. Hand-in is **Friday 27 November, 23:59**; the presentation is on **Wednesday 2 December**.

Phase	Dates	Tasks	Tools	Product
0 · Scope and setup	11–12 Nov	Fix fields and birth-year range; agree on roles; set up repository and paper template	GitHub, Overleaf (PNAS LaTeX template)	Scope note, shared repo, empty paper, data log
1 · Data collection	12–15 Nov	Query Wikidata for the scientist list and gender; download Wikipedia text and links; fetch OpenAlex impact data	Python, requests, SPARQL, MediaWiki API, OpenAlex API	nodes.csv, edges.csv, one text file per page, impact.csv
2 · Matching and cleaning	14–17 Nov	Link Wikidata IDs to OpenAlex authors (prefer ORCID / OpenAlex IDs); hand-check a sample of 100 matches; remove disambiguation and non-scientist pages	pandas	Clean dataset; match-accuracy figure for the Methods section
3 · Network analysis	16–20 Nov	Centrality vs. citations; within-bin gender shuffle; assortativity; communities	NetworkX, Louvain, NumPy	Figure 1, Figure 3, null-model results
4 · Text analysis	18–22 Nov	Personal vs. professional word lists; hand-label sentences; TF-IDF and Scattertext; page length	spaCy, scikit-learn, Scattertext	Figure 2, validated word lists
5 · Robustness	21–24 Nov	Repeat key results in a second field; different bin widths and seeds	Same as above	Robustness table (Methods / supplement)
6 · Writing	22–27 Nov	Write the paper around one story; slides; reflection; author contributions	Overleaf, Matplotlib	Paper (≤ 6 pages), 3 slides, reflection: **hand-in 27 Nov**
7 · Presentation	28 Nov–2 Dec	Rehearse the 3-minute talk; everyone prepares to explain every part	Slides PDF	Rehearsed talk and Q&A notes

Milestones and go/no-go points

Milestone	Date	Condition	If it fails
M1 · Data in hand	Sun 15 Nov	≥ 1,000 pages downloaded; ≥ 100 women matched to OpenAlex	Widen the birth-year range or add a field; if still short, switch to a fallback
M2 · Early null test	Wed 18 Nov (session 2)	First within-bin shuffle for RQ1 is run	If there is no gap, reframe the story around where the gap appears (citations vs. Wikipedia)
M3 · Three figures	Tue 24 Nov	Draft versions of Figures 1–3 exist	Cut the weakest analysis; three figures is the cap
M4 · Full draft	Wed 25 Nov (session 3)	Complete draft, read by all members	Use the session to cut and tighten, not to add analyses

6. Final products

- **Paper:** PNAS LaTeX template, at most six pages including figures and references: Abstract, Significance Statement, Introduction, Results, Discussion, Methods, References, and specific author contributions.
- **Slides:** three slides, three minutes, plain PDF (question · main result · what it means and its limits).
- **Reflection:** half a page on whether the course's “AI-drafted report” boxes changed how we read our own draft.
- **Code and data log** in the group repository, so every number in the paper can be traced.

7. Risks and how we handle them

Risk	Mitigation
Name ambiguity when matching to OpenAlex (two people with the same name)	Use ORCID or OpenAlex IDs from Wikidata first; hand-check a random sample and report the accuracy; drop unclear matches
OpenAlex covers older publications poorly	Restrict to scientists born 1920–1980; test sensitivity to the range
Wikidata gender is mostly binary and sometimes missing	Say so explicitly in Methods; analyse only recorded values; report how many were missing
Too few women for stable comparisons in high-citation bins	Use wider bins at the top; report counts per bin; avoid claims about thin bins
Prior research already found Wikipedia gender gaps	Use it in the literature section; our contribution is controlling for impact and combining network and text
Running out of time	Milestones M1–M4; three figures maximum; no new analyses after 24 November

8. Fallback projects

- **Reddit gateway:** manosphere dataset (Ribeiro et al., Zenodo) combined with fitness/self-improvement subreddits from Arctic Shift; users linked by shared threads.
- **YouTube audiences:** channel–commenter network along a fitness-to-manosphere gradient, if the YouTube Data API is confirmed to be acceptable.

9. Roles (to fill in)

Member	Main responsibility	Also owns
Member A	Data collection and matching (Phases 1–2)	Data log, Methods: data
Member B	Network analysis and null models (Phase 3)	Figures 1 and 3
Member C	Text analysis and hand labeling (Phase 4)	Figure 2, reflection
Everyone	Writing, robustness checks, slides	Being able to explain every part

The paper's author contributions must name concrete tasks, so we keep this table updated as the work shifts.